

# Engineering Standards Manual

Meridian Quality Systems  
Internal Engineering Documentation · Revision 3.4 · Expert Guide

## About This Document

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| Difficulty             | ★★★★ Expert                                                                                                   |
| Best for               | Full Pipeline Stress-Testing, Conflicting/Updated Policy Resolution, OCR-Noise Tolerance, Multi-Hop Synthesis |
| Approximate Chunks     | 95                                                                                                            |
| Estimated Embeddings   | 95                                                                                                            |
| Recommended RAG Stages | Parsing → Chunking → Retrieval → Generation → Evaluation                                                      |

This document intentionally contains nearly every real-world problem a production RAG pipeline must handle. Some questions in this document have no answer anywhere in the text.

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## 1. Document Control

| Revision | Date       | Author                     | Summary                                                                               |
|----------|------------|----------------------------|---------------------------------------------------------------------------------------|
| 1.0      | 2021-03-01 | J. Alvarez, Eng. Standards | Initial release                                                                       |
| 2.0      | 2022-07-15 | J. Alvarez, Eng. Standards | Added Incident Escalation Procedure                                                   |
| 3.0      | 2023-11-02 | R. Okafor, Eng. Standards  | Added Data Retention Policy                                                           |
| 3.4      | 2024-05-20 | R. Okafor, Eng. Standards  | Added Retry Policy and Data Retention addenda; superseded relevant 2022/2023 sections |

**NOTE:** This manual applies to all Meridian Quality Systems engineering teams, including teams responsible for FalconTest, APIVerify, InsightIQ, MockBridge, and BuildFlow. Where this manual conflicts with a product-specific guide, the most recently dated policy in this manual takes precedence for anything within this manual's scope (organizational process, not product default behavior).

## 2. Purpose and Scope

This manual defines mandatory engineering standards for all MQS engineering teams. It covers retry policy at the organizational level, environment access, incident escalation, code review, data retention, and build pipeline standards. It does not cover product-specific configuration defaults, which remain documented in each product's own guide (see References).

### 3. Definitions and Abbreviations

| Term  | Definition                                                     |
|-------|----------------------------------------------------------------|
| MQS   | Meridian Quality Systems                                       |
| SEV-1 | Severity 1 incident: production-impacting, customer-facing     |
| SEV-2 | Severity 2 incident: production-impacting, not customer-facing |
| RPO   | Recovery Point Objective                                       |
| RTO   | Recovery Time Objective                                        |
| PII   | Personally Identifiable Information                            |

Full abbreviation list continues in Appendix A.

### 4. Retry Policy (2022 Edition)

*Effective 2022-07-15. See Section 9 for the 2024 addendum, which partially supersedes this section.*

All MQS testing tools, including FalconTest's Smart Retry engine, must default to a global retry limit of **3 attempts** for any automated test run, regardless of execution mode. This limit exists to prevent runaway retry loops from consuming shared ClusterGrid capacity during periods of environment instability.

Teams may not configure a retry limit higher than 3 without written approval from the Engineering Standards team, filed as an exception request.

## 5. Environment Access Standards

Access to shared testing environments (staging, pre-production, and any environment connected to ClusterGrid) is granted through the standard access request process. Requests are reviewed within two business days.

| Environment Tier       | Approval Required                     | Access Duration    |
|------------------------|---------------------------------------|--------------------|
| Development            | Team lead                             | Indefinite         |
| Staging                | Team lead                             | 90 days, renewable |
| Pre-Production         | Engineering Standards team            | 30 days, renewable |
| Production (read-only) | Engineering Standards team + Security | 14 days, renewable |

**WARNING:** Production write access is never granted through this process. See the separate Production Access Policy (not part of this manual) for that process.

## 6. Incident Escalation Procedure

When a production-impacting issue is detected in any MQS-operated system, including InsightIQ Cloud deployments, the following escalation procedure applies.

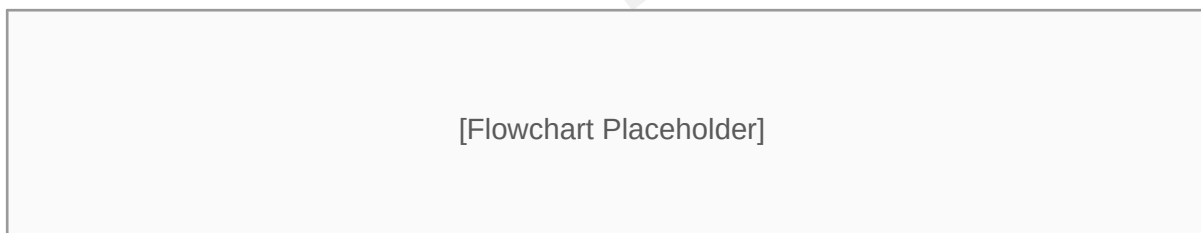


Figure 4: Incident escalation flow from detection to resolution.

- 1 **Detect** — the issue is identified, either through automated monitoring or a manual report.
- 2 **Classify** — the on-call engineer classifies the issue as SEV-1 or SEV-2 using the definitions in Section 3.
- 3 **Escalate** — SEV-1 incidents page the on-call Engineering Standards representative immediately; SEV-2 incidents are logged and reviewed at the next business-hours triage meeting.
- 4 **Resolve** — the incident is resolved and a post-incident review is scheduled within 5 business days for any SEV-1 incident.

See Appendix C for a duplicate copy of this procedure that was merged into this manual from a legacy wiki page during the 2023 documentation migration; Appendix C's wording differs slightly but describes the same procedure, and the version in this section (Section 6) is the authoritative one.

## 7. Code Review Requirements

All code changes to MQS-operated systems require review before merging. The following table describes review requirements by change category, organized as a nested structure: top-level rows are change categories, and each category groups two related sub-requirements.

| Change Category                     | Requirement                                  | Minimum Reviewers |
|-------------------------------------|----------------------------------------------|-------------------|
| <b>Standard change</b>              |                                              |                   |
| — Application code                  | Peer review required                         | 1                 |
| — Configuration change              | Peer review required                         | 1                 |
| <b>Sensitive change</b>             |                                              |                   |
| — Authentication/authorization code | Peer review + Security sign-off              | 2                 |
| — Production data migration         | Peer review + Engineering Standards sign-off | 2                 |
| <b>Emergency change</b>             |                                              |                   |
| — SEV-1 hotfix                      | Post-hoc review within 24 hours              | 1 (post-hoc)      |
| — SEV-1 configuration rollback      | Post-hoc review within 24 hours              | 1 (post-hoc)      |

### 7.1 Reviewer Checklist

The following two-column checklist summarizes what a reviewer must confirm before approving any Sensitive-change category pull request; both columns apply to every sensitive change reviewed.

#### Column A — Code Quality

Tests added or updated for the change.

No hardcoded credentials or secrets.

Follows the team's existing code style.

No unrelated changes bundled into the same pull request.

#### Column B — Risk Assessment

Rollback plan documented in the pull request description.

Blast radius identified (which systems/customers are affected).

Security sign-off obtained if authentication/authorization code is touched.

Engineering Standards sign-off obtained if a production data migration is involved.

## 8. Data Retention Policy (2022 Edition)

*Effective 2023-11-02. See Section 10 for the 2024 addendum, which partially supersedes this section.*

All test execution logs generated by FalconTest, APIVerify, and InsightIQ must be retained for a minimum of **30 days** from the date of generation, after which they may be deleted or archived at the owning team's discretion.

This retention requirement applies to logs only. Contract Baselines (APIVerify), Test Runner configuration, and other non-log artifacts are not subject to this policy and should be retained according to normal source-control practices.

## 9. Retry Policy — Addendum (2024)

*Effective 2024-05-20. This addendum partially supersedes Section 4.*

Following a review of ClusterGrid capacity incidents in early 2024, the organizational default retry limit is revised as follows:

- For **scheduled or unattended pipelines** (including any FalconTest suite running in Adaptive mode as part of a nightly or scheduled job), the default retry limit is increased to **5 attempts**.
- For **interactive or CI-gating pipelines** (any pipeline that blocks a merge or deployment and is expected to return a result while a person is actively waiting), the retry limit **remains capped at 3 attempts**, unchanged from Section 4.

Teams should note that FalconTest's own documentation (see the FalconTest Automation Framework Guide, Section 5.2) describes Smart Retry's default `max_attempts` as 3 at the product-configuration level. This organizational policy does not change that product default; it requires teams running scheduled/unattended FalconTest suites to explicitly override `max_attempts` to 5 in their suite's `falcontest.yaml`, consistent with this addendum.

## 10. Data Retention Policy — Addendum (2024)

*Effective 2024-05-20. This addendum partially supersedes Section 8.*

For any test execution log associated with a SEV-1 incident (as defined in Section 3 and escalated per Section 6), retention is extended from the standard 30 days to **90 days**, to ensure logs remain available throughout any post-incident review and follow-up audit.

All other test execution logs not associated with a SEV-1 incident remain subject to the original 30-day retention policy defined in Section 8. This addendum does not change retention for APIVerify Contract Baselines, Performance Baselines, or InsightIQ evaluation datasets, none of which are classified as “test execution logs” under this policy.

## 11. Deprecated Build Pipeline Procedure

*Deprecated as of 2023-09-01. Retained here for historical reference only. Do not follow this procedure for new pipelines — see Section 12 for the current procedure.*

*[Note: this section was migrated from a legacy scanned PDF during the 2022 documentation consolidation and retains some artifacts from that OCR pass.]*

The legacy build pipeline configuration required teams to manually register each new service with BuildFlow using the `buildflow register` command, then manually attach a ClusterGrid execution profile before the pipeline could run any FalconTest suites as part of a merge gate.

```
buildflow register --service my-service --team my-team  
buildflow attach-profile --service my-service --profile clustergrid-default
```

This procedure required a minimum of two manual steps per service and was a common source of onboarding delays for new teams, since the `attach-profile` step had no validation and silently failed if the profile name was misspelled.



## 12. Current Build Pipeline Procedure

*Effective 2023-09-01, replacing the procedure in Section 11.*

BuildFlow now auto-detects and registers new services on first push, using a `buildflow.yaml` manifest committed to the service's repository. No manual `buildflow register` or `attach-profile` step is required.

```
service: my-service
team: my-team
cluster_profile: clustergrid-default
falctest_gate: true
apiverify_gate: true
```

If `falctest_gate` and `apiverify_gate` are both set to `true`, BuildFlow will block a merge unless the relevant FalconTest suite (Passive mode, per the FalconTest Automation Framework Guide, Section 4.1) and APIVerify Endpoint Groups both pass.

## 13. Multi-Region Deployment Standards

*The following section is formatted as two columns in the PDF rendering, consistent with a print-style standards appendix; both columns should be read left-to-right, top-to-bottom, as a single continuous requirement, not as two independent lists.*

### Deployment Requirements

Every production service must be deployed to a minimum of two MQS-operated regions.

ClusterGrid capacity must be reserved in both regions before a service is declared production-ready.

Failover between regions must be tested at least once per quarter, with results logged per the retention requirements in Section 8/10.

### Verification Requirements

APIVerify Endpoint Groups for any multi-region service must be run against both regions independently; a passing result in one region does not imply a passing result in the other.

Drift Detection (see the AI Testing & RAG Playbook, Section 6) findings must be reviewed per-region, since regional configuration differences are a common source of region-specific drift that a single-region check would miss entirely.

## Appendix D — Sample SEV-1 Incident Log (2023–2024)

The following is a sample excerpt of the SEV-1 incident log referenced by the retention requirements in Section 10. It is included as a long-table example; the real log is maintained in InsightIQ's incident tracking index, not in this manual.

| Incident ID | Date       | Product     | Region  | Summary                                                            | Duration | Status   |
|-------------|------------|-------------|---------|--------------------------------------------------------------------|----------|----------|
| INC-2301    | 2023-01-14 | APIVerify   | us-east | Drift Detection false-positive storm after schema migration        | 3.2h     | Resolved |
| INC-2308    | 2023-02-02 | FalconTest  | us-west | ClusterGrid capacity exhaustion during nightly Adaptive run        | 5.1h     | Resolved |
| INC-2314    | 2023-02-19 | BuildFlow   | us-east | Merge gate stuck due to attach-profile misconfiguration            | 1.4h     | Resolved |
| INC-2322    | 2023-03-08 | InsightIQ   | eu-west | Embedding model mismatch after partial re-index                    | 2.0h     | Resolved |
| INC-2331    | 2023-04-11 | APIVerify   | eu-west | Baseline Sync auto-schedule enabled in error for Strict Mode group | 0.8h     | Resolved |
| INC-2345    | 2023-05-06 | FalconTest  | us-east | Reporting Cluster backlog delayed report publication               | 4.6h     | Resolved |
| INC-2359    | 2023-06-21 | ClusterGrid | us-west | Worker node health-check flapping under load                       | 6.3h     | Resolved |
| INC-2367    | 2023-07-30 | APIVerify   | us-east | Contract Baseline approved in error, reverted same day             | 0.5h     | Resolved |
| INC-2378    | 2023-08-17 | InsightIQ   | eu-west | Re-Ranking latency spike on InsightIQ Cloud                        | 2.9h     | Resolved |
| INC-2389    | 2023-09-25 | BuildFlow   | us-west | Legacy attach-profile step silently failed for new team            | 3.7h     | Resolved |
| INC-2402    | 2023-11-09 | FalconTest  | us-east | Execution Cluster misrouted jobs after regional failover test      | 5.5h     | Resolved |
| INC-2415    | 2023-12-18 | APIVerify   | eu-west | Drift severity misclassified as Info instead of Critical           | 1.9h     | Resolved |
| INC-2428    | 2024-01-22 | ClusterGrid | us-west | Capacity reservation conflict between two regions                  | 4.2h     | Resolved |
| INC-2441    | 2024-02-14 | InsightIQ   | eu-west | ModelMismatchError surfaced in production query path               | 1.1h     | Resolved |
| INC-2456    | 2024-03-30 | FalconTest  | us-east | Smart Retry loop exceeded pre-addendum retry limit                 | 3.0h     | Resolved |
| INC-2470    | 2024-05-21 | APIVerify   | us-west | Post-addendum retention gap identified for one log stream          | 0.6h     | Resolved |

## Appendix A — Abbreviations

| Abbreviation | Meaning                             |
|--------------|-------------------------------------|
| MQS          | Meridian Quality Systems            |
| SEV-1        | Severity 1 incident                 |
| SEV-2        | Severity 2 incident                 |
| RPO          | Recovery Point Objective            |
| RTO          | Recovery Time Objective             |
| PII          | Personally Identifiable Information |
| CI           | Continuous Integration              |
| CD           | Continuous Deployment               |
| p95          | 95th percentile                     |

## Appendix B — Cross-Reference Index

| This Manual  | Referenced Document                   | Section                                                                                              |
|--------------|---------------------------------------|------------------------------------------------------------------------------------------------------|
| Section 4, 9 | FalconTest Automation Framework Guide | Section 5.2 (Smart Retry)                                                                            |
| Section 6    | FalconTest Automation Framework Guide | Section 2 (Execution/Reporting Cluster)                                                              |
| Section 12   | FalconTest Automation Framework Guide | Section 4.1 (Passive Mode)                                                                           |
| Section 13   | APIVerify Integration Handbook        | Section 2.2 (Endpoint Groups)                                                                        |
| Section 13   | AI Testing & RAG Playbook             | Section 6 (Drift Detection reference is to Playbook Section 4, cross-linked from Playbook Section 6) |

## Appendix C — Incident Escalation Procedure (duplicate, see note)

*This appendix is a near-duplicate of Section 6, preserved from a legacy wiki page merged into this manual in 2023. The wording below differs slightly from Section 6 but describes the same underlying procedure. Section 6 is authoritative; this appendix exists only because the migration process did not remove the original page content.*

When a production-affecting problem is found in an MQS-run system, staff should follow these steps: first, the problem is identified via monitoring tooling or a manual report from a team member; second, the on-call engineer determines whether the problem is SEV-1 or SEV-2 severity; third, SEV-1 problems result in an immediate page to the on-call Engineering Standards rep, while SEV-2 problems get logged for the next triage meeting; fourth, once resolved, a post-incident review is scheduled for SEV-1 issues within five business days of resolution.

## Glossary

| Term                 | Definition                                                                               |
|----------------------|------------------------------------------------------------------------------------------|
| Retry limit          | The maximum number of times an automated test is re-attempted after failure              |
| Contract Baseline    | (See APIVerify Integration Handbook, Section 2.3)                                        |
| Test execution log   | A log generated by FalconTest, APIVerify, or InsightIQ during a test or verification run |
| Post-incident review | A structured review conducted after a SEV-1 incident is resolved                         |
| Blast radius         | The set of systems or customers affected by a given change or incident                   |

## References

- 1 FalconTest Automation Framework Guide — this benchmark suite, Document 1.
- 2 APIVerify Integration Handbook — this benchmark suite, Document 2.
- 3 AI Testing & RAG Playbook — this benchmark suite, Document 3.
- 4 Production Access Policy — referenced in Section 5, not part of this benchmark suite.

## Questions to Try

The following questions are provided to help you explore this document using your own RAG pipeline. No answers are provided here — try retrieving and generating answers yourself. Some of these questions intentionally have no answer anywhere in this document.

- 1 What is the current organizational retry limit for a FalconTest suite running in Adaptive mode as part of a nightly scheduled job?
- 2 How long must a test execution log associated with a SEV-1 incident be retained?
- 3 Compare the 2022 and 2024 Retry Policy sections. What changed, and what stayed the same?
- 4 A new production service needs to go through code review, get deployed to two regions, and pass an incident-escalation drill. Which sections of this manual apply, and in what order would a team typically use them?
- 5 How many reviewers are required for a production data migration change?
- 6 What is “the Incident Escalation Procedure” in this manual — the version in Section 6, or the version in Appendix C?
- 7 What encryption standard does this manual require for data at rest?
- 8 What was required to register a new service with BuildFlow under the deprecated build pipeline procedure, and what replaced it?
- 9 Does the 2024 Data Retention addendum change the retention period for APIVerify Contract Baselines?
- 10 Why does this manual say the 2024 Retry Policy addendum does not contradict FalconTest's own documented default of 3 retry attempts?